

Kartikeswar Rana

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Profile

Motivated M.Sc. Physics student with a strong foundation in physics, mathematics and a disciplined approach to continuous learning. Focused on building clear concepts, analytical thinking, and problem-solving abilities. Experienced in teaching higher secondary mathematics with emphasis on simplifying complex topics for better understanding through structured study. Committed to consistently strengthening both academic knowledge and teaching effectiveness for consistent personal and student growth.

Personal Details

- Date of Birth : 17th November 2003
- Father's Name : Sudarshan Rana
- Mother's Name : Kamalini Rana
- Nationality : Indian
- Address : At/Po - Darada, PS - Basta, Balasore, Odisha – 756022

Education

Qualification	Institution	Year	Result
M.Sc. in Physics	Fakir Mohan University, Odisha	2024 – 2026	7.19/10 OGPA
B.Sc. in Physics	Sulagna +3 Science Degree College	2021 – 2024	78% (Distinction)
Class XII	Meghasan Higher Secondary School	2019 – 2021	59%
Class X	Durgadevi Nodal High School	2019	56%

M.Sc. Thesis / Project

Exploring Primordial Black Hole Evolution Using Python Interpreter

Supervisor: Dr. Bibekananda Nayak, Fakir Mohan University

Poster Presentations

A Study on Primordial Black Hole Dynamics using Python	Feb 2026
<i>National Seminar on Quantum Physics to Quantum Technology and 42nd Annual Convention of OPS, Berhampur University, Odisha</i>	
Time Evolution of Primordial Black Holes: A Numerical Approach	Feb 2026
<i>International Conference on Recent Advances in Physics, VSSUT, Sambalpur, Odisha</i>	
The Lifetime of Primordial Black Holes Under Accretion and Evaporation	Mar 2026
<i>Frontier Developments in Physics and Atmospheric Sciences, Fakir Mohan University, Balasore, Odisha</i>	

Achievements

- Secured All Odisha Rank 53 in CPET-2024 (Physics)
- Secured All Odisha Rank 08 in CPET-2024 (Applied Physics)

Skills

- **Programming:** Python
- **Operating Systems:** Linux, Windows
- **Tools:** $\text{\LaTeX}$
- **Online Teaching Tools:** Teachmint, Google Meet
- **Office Tools:** Excel, PowerPoint, Word
- **Areas:** Mathematical Physics, Computational Simulation, Quantum Mechanics, Cosmology

Experience

Mathematics Tutor at Utkala Physics

April 2025 – Present

- Teaching Class 11 and 12 students through online and offline mode
- Developed question sets, formatted academic materials, and created presentations using $\text{\LaTeX}$ and Microsoft Office, ensuring clarity in mathematical content and layout
- Focus on conceptual clarity and problem-solving

Certifications

- **NPTEL (IIT Kanpur)** – Applied Numerical Methods (84%, Elite) (Jul–Oct 2025)
- **ISRO IIRS** – Remote Sensing & Numerical Ocean Modelling (Aug 2025)
- **NIT Rourkela** – Multiscale Modelling in Biophysics Workshop (Jul 2025)
- **IIRA** – Data Analysis using Excel (Nov 2025)
- **IIRA** – Literature Review & Plagiarism Management (Zotero) (Oct–Nov 2025)

Interests

- Scientific computing and simulation (Python)
- Reading non-fictional books
- Exploring data analysis and visualization techniques
- Creating educational content and notes for physics and mathematics
- Learning and experimenting with Linux tools and workflows

Declaration

I hereby declare that the above information is true to the best of my knowledge and belief.

Date:

Place:

Signature